

Shortcut Learning in Large-Scale Models

Executive Summary:

- Large language models often still rely on trivial cues (shortcuts) rather than true semantics. For example, Tang et al. (2023) found that LLMs act like “lazy learners” exploiting simple prompt shortcuts, and that *larger* models paradoxically rely on them **more** ¹ ². Yuan et al. (2024) show this concretely: when tested on NLI tasks with synthetic shortcut patterns, models’ accuracy can drop by 40% or more ³, even though they perform well on standard data. Prompting strategy matters: chain-of-thought prompting reduces shortcut use, while simple few-shot prompts can backfire ⁴. However, models remain overconfident and give poor explanations on shortcut examples ⁵. In practice, tiny changes can break LLMs – for instance, simply permuting the answer choices in a multiple-choice question dramatically cuts accuracy ⁶.

- Training methods also affect shortcuts. Marek et al. (2025) showed that *very* small batch sizes (even 1) can train LMs stably if Adam’s decay is scaled properly. Small-batch training proved **more robust** to hyperparameter choices and just as efficient per FLOP as large batches ⁷. In fact, they could train an LM with vanilla SGD (no momentum) by using tiny batches. Their advice: favor genuine small-batch training (with adjusted hyperparams) over naive gradient-accumulation.
- This report surveys these findings and related literature (2020–2026). We review major benchmarks (e.g. **HANS**, **WILDS/Waterbirds**, sentiment/NLI stress tests), evaluation metrics (accuracy drop, *worst-group* accuracy ⁸), and new LLM-specific scores like Semantic Fidelity and Internal Consistency ⁹, and common failure modes (e.g. distraction by irrelevant features, logical fallacies in explanations). We analyze why shortcuts arise – data biases, model inductive biases, optimization choices, pretraining effects – and highlight empirical case studies (image vs. language tasks) showing how simple spurious cues cause errors. We then survey mitigation strategies: data-centric (counterfactual augmentation, rebalancing), training-based (IRM/DRO, optimizer schedules, batch size tweaks), architectural and regularization methods (e.g. encouraging shape-bias ¹⁰), and prompt engineering (chain-of-thought, calibration). Finally, we give practical experimental recommendations (which datasets, metrics, hyperparameters, and ablations to use) and outline open problems in making LLMs robust to shortcuts.

timeline

title Shortcut Learning Research Milestones (2018–2025)

2018-01-01 : Naik et al. (COLING’18) introduce NLI negation stress tests ¹¹.

2019-01-01 : McCoy et al. (EMNLP’19) release HANS dataset to probe NLI heuristics (lexical overlap, subsequence, etc.) ¹².

2020-01-01 : Arjovsky et al. propose Invariant Risk Minimization (IRM) to combat spurious features; Koh et al. launch WILDS benchmark (Waterbirds, CelebA, etc.) ¹³ ⁸.

2021-01-01 : Surveys of spurious correlations appear (e.g. Ye et al. 2024) summarizing datasets/metrics; wider adoption of worst-group accuracy ⁸.

2023-01-01 : Tang et al. (ACL Findings’23) demonstrate LLM “lazy learning”

shortcuts; Zong et al. (arXiv'23) expose LLM vulnerability via answer-permutation attacks ² ¹⁴ .

2024-01-01 : Yuan et al. (EMNLP'24) release the "Shortcut Suite" of NLI tests with new metrics ³ ⁴ ; reviews (e.g. Steinmann et al.) survey shortcut learning.

2025-01-01 : Marek et al. (NeurIPS'25) show small-batch (even 1) SGD works for LM training

⁷ ; Pathak et al. (Front. AI, 2026) publish an expansive "Clever Hans" effect review.

Primary Studies

Yuan et al. (EMNLP 2024) – "*Do LLMs Overcome Shortcut Learning?*" introduced a **Shortcut Suite** of NLI test sets designed to isolate six textual shortcuts (Table 1 in [39†L296-L304]), e.g. syntactic heuristics and irrelevant style cues. They also define new metrics (Semantic Fidelity, Internal Consistency, Explanation Quality, etc.) to evaluate LLM explanations ⁹ . Their experiments with multiple LLMs and prompts found: (1) LLMs do rely on shortcuts – e.g. accuracy often drops >40% on constituent/negation shortcuts ³ ; (2) *larger* LLMs are **more** prone to these shortcuts under zero/few-shot prompts ⁴ , a "reverse scaling" effect; (3) chain-of-thought prompting significantly reduces shortcut use, whereas few-shot prompts surprisingly hurt (LLMs did better zero-shot than few-shot) ⁴ ; (4) LLMs are often overconfident on the shortcut sets (low "confidence score" despite many mistakes) ⁹ ; (5) generated explanations are lower-quality on these cases, exhibiting "distraction" and "logical fallacies" in place of real reasoning ⁴ . In short, Yuan *et al.* show that even GPT-3/GPT-4-level models can fail dramatically on contrived shortcut cases, and they quantify it with rigorous new metrics.

Marek et al. (NeurIPS 2025) – "*Small Batch Size Training for Language Models*" challenges the conventional wisdom that large LMs require large batches or Adam. They derive a simple rule: when scaling Adam to small batches, keep the *half-life* of its second moment constant (rather than its usual decay rate). With this trick, they show training is stable even with batch size = 1. Key findings: **(1)** Very small batches can train stably; **(2)** small-batch training is more robust to hyperparameter choice; **(3)** on a per-FLOP basis, small batches can match or outperform large batches; **(4)** one can even use *vanilla* SGD (no momentum, no adaptive state) to train an LM with these settings ⁷ . They recommend picking the largest batch size that fits memory (or choosing very small batches with scaled Adam), and discourage gradient-accumulation (which wastes the benefits of small-batch noise). For example, they show a LoRA-finetuned model using batch=1 SGD achieves full-finetune performance with only LoRA memory cost ⁷ . This work suggests simple training-interventions (batch & optimizer tuning) can improve LM learning dynamics, which may indirectly reduce overfitting to shortcuts by injecting noise.

Dataset Benchmarks and Metrics

Datasets/Benchmarks: Various benchmarks have been devised to *reveal* shortcut learning in ML models. In vision, the WILDS collection (Koh et al. 2020) is widely used: e.g. *Waterbirds* (bird species classification with correlated land/water backgrounds) and *CelebA* subsets (e.g. Gender vs. Blondes; Lipstick vs. gender). These have known spurious cues (background, facial attributes) ¹⁵ . Other vision tests include Colored MNIST (digit color spurious), corrupted CIFAR, and stylized ImageNet (Geirhos et al. 2019) which forces reliance on shape over texture ¹⁰ . In NLP, common shortcuts arise in Natural Language Inference (NLI)

datasets: e.g. HANS (McCoy et al. 2019) targets lexical overlap, subsequence, and constituent heuristics ¹². Stress tests for NLI (Naik et al. 2018) create examples with negation or lexical swaps – models often fail these ¹¹. Sentiment classification (SST, IMDB) and QA datasets also harbor artifacts (like certain keywords or negation patterns).

For **LLMs**, specialized tests are emerging. Tang et al. injected trivial *prompt triggers* into SST-2, MR, etc., while Zong et al. (2023) designed *ShortcutQA* by permuting MCQA answer sets. For example, a simple MCQA permutation test uses MMLU or VQA questions with shuffled choices – models should be invariant, but fail badly ⁶. These and similar tests (e.g. swapping tokens in questions, using odd distractors) expose LLM shortcut tendencies.

Metrics: Common evaluation metrics include: - **Accuracy drop / worst-group accuracy:** Compare performance on a *shortcut-based* test set vs. standard set. The *worst-group accuracy* (minimum accuracy across groups defined by spurious features) is often reported ⁸. For example, one measures the accuracy on cases that *conflict* with training biases (e.g. waterbirds on land) ¹⁶. - **Consistency and Explanation scores:** Yuan et al. introduce *Semantic Fidelity* (do explanations match answers?) and *Internal Consistency* (is each explanation self-consistent?) to gauge if the model “really” understands its answers ⁹.

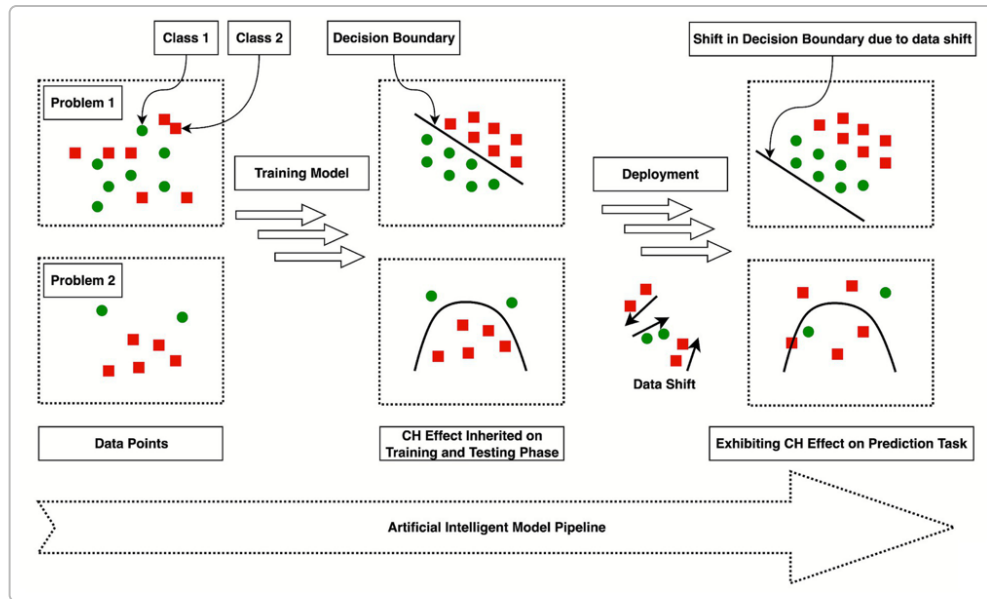
- **Confidence/Calibration:** Overconfidence on spurious examples can be measured (e.g. a high predicted probability for a wrong shortcut answer). Yuan’s “Confidence Score” checks if model’s stated confidence matches actual correctness.

- **Behavioral tests:** For LLMs, one tracks failure modes qualitatively (e.g. “distraction” where model fixates on irrelevant phrase). For simpler models, one might visualize decision boundaries shifting under bias.

Typical **failure modes** include: high performance on training biases (even when the true features are removed), but nearly 0 performance when those cues are absent ³ ¹⁶. Models may give plausible-sounding but wrong rationales, or quickly flip answers when surface patterns change. For example, in Tang et al.’s sentiment task, inserting a common word at the end caused GPT2-large to drop ~40% accuracy, revealing it was using that word as a shortcut ¹⁷. In Zong et al., shuffling MCQA answers flipped the model’s choice (see below). In vision, a model might learn that “cow on grass = cow” and fail on a “cow on road” image. Any such behavior can be detected by targeted test sets or counterfactual examples.

Mechanisms of Shortcut Emergence

Shortcut learning arises from a confluence of data, model, and training factors. Pathak et al. (2026) summarize key causes: (1) **Data artifacts/biases** – non-causal features (background, token patterns, genre tags, etc.) that correlate with labels in training ¹⁸; (2) **Lack of causal supervision** – models are not told which features are “true”, so they take any predictive shortcut ¹⁸; (3) **Imbalanced or confounded data** – majority groups dominate so the model ignores minority patterns ¹⁸; (4) **Insufficient evaluation** – standard i.i.d. test splits hide the problem ¹⁸; (5) **Model inductive bias** – deep nets preferentially learn “simple” features.



For example, Figure 1 (from Pathak et al.) illustrates a “Clever Hans” scenario: the training data (left) contains spurious cues (colored blocks) that allow a simple decision boundary. The model learns this boundary (middle). Under a data shift (right), that boundary no longer aligns with the true concept, leading to failure. Empirically, Hermann et al. (2024) demonstrated that this is inherent to depth: in a toy task with an “easy” and a “true” feature, a linear classifier learned the true feature, but adding just one hidden layer made the network prefer the easy shortcut ¹⁹. In other words, *deep nonlinear architectures inherently favor readily-available predictive signals*. Similarly, Geirhos et al. (2019) found that CNNs trained on ImageNet are strongly texture-biased ¹⁰ – they pick up on local patterns rather than global shape because textures are easier to exploit, a classic shortcut.

Optimization and architecture matter too. Overparameterized models can memorize dataset idiosyncrasies as well as true concepts. Pathak et al. note that choices like batch size and optimizer can tilt the balance between signal and noise. For instance, Marek et al. (2025) show that small-batch SGD injects noise that can discourage overfitting to spurious features, yielding different solutions than large-batch Adam ⁷. Conversely, gradient accumulators (large effective batches) may allow the model to converge on the simplest fit to data. Pretraining on massive corpora may introduce its own biases (for example, a language model might see certain word co-occurrences very often), and fine-tuning on biased data can then amplify these patterns. In summary, shortcuts arise whenever **model capacity, data bias, and training dynamics** align to make a spurious cue more attractive than the true underlying rule ¹⁸ ¹⁹.

flowchart LR

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Data["Data biases (spurious features)"] --> ModelLearn["Model learns shortcut pattern"]
Arch["Deep architecture (capacity)"] --> ModelLearn
Opt["Training (optimizer, batch, etc.)"] --> ModelLearn
ModelLearn --> Fail["Generalization failure on new data"]
Shift["Distribution shift"] --> Fail

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Empirical Case Studies

Recent experiments illustrate how shortcuts propagate from small tasks to large models:

- **Tang et al. (2023)** – *In-context LLM shortcuts*. They tested GPT-2 and OPT models on sentiment tasks (SST2, MR) by inserting trivial triggers (like the word “movie” or a simple phrase) into prompts. GPT2-large’s accuracy on MR fell from ~63% to 22% (a ~41.5% drop) when the word “movie” was appended ¹⁷, whereas GPT2-base’s accuracy barely changed. Overall, larger models showed **much bigger** performance drops: e.g. GPT2-large averaged a 33.7% drop versus only 1.0% for GPT2-base ². This “inverse scaling” implies GPT2-large was relying on the shortcut. They also found that placing the trigger at the end of the input had the strongest effect, and that sentence-level triggers (“This is a shortcut.”) caused near-total failure. These results directly demonstrate that even without fine-tuning, larger LLMs can latch onto shallow patterns in the prompt ¹ ².
- **Zong et al. (2023)** – *Permutation Attack on MCQA*. They considered multiple-choice QA tasks (like MMLU or VQA). Although answer order should not matter, they found LLMs are sensitive to it. For example, LLaMA-2 13B’s accuracy on MMLU dropped by ~33.9% after randomly shuffling the answer options ¹⁴. In one illustrative case (Figure 1 below), the model originally answered “Two women” correctly; after a random permutation of the options, the model instead answered “An old man” incorrectly ⁶ [63†]. This shows the model had learned to associate certain answers with certain positions, a trivial shortcut. (Higher distractor count or calibration did not solve the issue.)

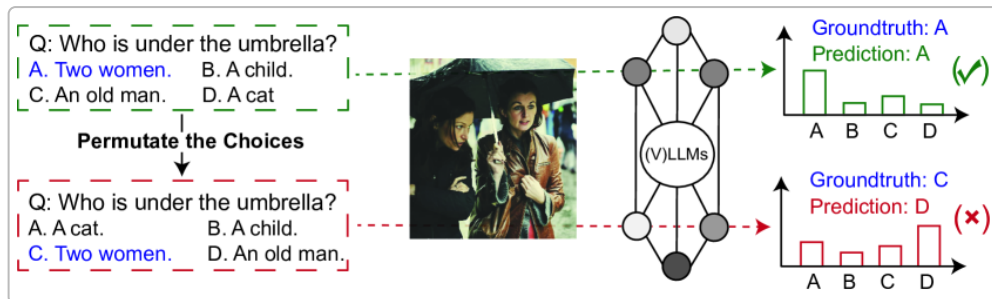


Figure: An example from Zong et al. – before (top) and after (bottom) randomly permuting multiple-choice answers. The LLM’s output distribution shifts and it picks the wrong answer after the shuffle ⁶.

- **Chan et al. (2022)** – *Vision models and model size*. They studied how model scale affects robustness on spurious datasets (Waterbirds, CelebA). Interestingly, under standard ERM training, **larger** models were **more** robust: e.g. a pre-trained ResNet152 achieved ~18.4% *lower* worst-group error than ResNet18 on Waterbirds ¹⁶. In fact, worst-group accuracy *improved* with size when using pretrained models. This contrasts with the LLM findings: it suggests that with sufficient data and gradient updates, bigger vision models may actually learn the causal features more reliably ¹⁶. (However, when rare examples are extremely few, any model struggles – Chan et al. also note that naive ERM still fails worst-case regardless of size.)
- **Hermann et al. (2024)** – *Synthetic linear vs. deep*. On a toy 2-feature task, a linear model (no hidden layers) typically learned the “true” feature, but adding a single hidden layer caused the network to use the easy (but spurious) feature instead ¹⁹. This mathematically supports that shortcuts are a **nonlinear** phenomenon: deep nets gravitate to the quick fix.

These case studies show shortcuts appear in both small and large models and across domains. Simple pattern-based attacks (triggers, permutations) that break a small model also break large LMs. Conversely, larger models trained fully can sometimes overcome some biases (as in Chan et al.), so the interplay of scale, data, and training is subtle.

Mitigation Strategies

Researchers have proposed many ways to reduce shortcut reliance. Broadly these include **data-based**, **training-based**, and **model-based** approaches:

- **Data interventions:** Balance or de-bias the training data. For example, counterfactual data augmentation introduces examples that *break* the spurious correlation. In NLI, one can add pairs where the usual heuristic is flipped (e.g. swap words, negate hypotheses) ¹¹. In vision, generative “style transfer” (e.g. training on Stylized-ImageNet) forces the model to learn shape instead of texture ¹⁰. Datasets like WILDS encourage balanced sampling across groups (so backgrounds or attributes aren't predictive). Adversarial filtering can remove easy cues from the data. In practice, increasing minority-group examples (or downsampling the majority bias group) directly raises worst-group accuracy ¹⁶. These interventions aim to make spurious features less statistically predictive.
- **Training regimens:** Modify the learning process. A key example is **batch size**: Marek et al. (2025) show that tiny batches (with scaled Adam or SGD) can improve stability ⁷. Gradient noise (from small-batch SGD) may help the model avoid sharp fitting to spurious quirks. Other optimizer choices matter too: some work suggests SGD can yield flatter minima than Adam, potentially reducing overfitting. Curriculum learning (introducing “hard” OOD examples earlier) and early stopping on held-out “shortcut tests” can also prevent overlearning. Importantly, as Yuan et al. found, prompting strategies like **chain-of-thought** effectively force the model to use multi-step reasoning, which reduces shortcut usage ⁴ – this is a kind of “training at inference” hack. Similarly, techniques like prompt ensembling or self-consistency averaging can mitigate sensitivity to any single shortcut.
- **Regularization and robust objectives:** One can bake robustness into the loss. Classic methods include **Group DRO** (distributional robust optimization), which upweights examples from minor groups so the model optimizes worst-case accuracy. Sagawa et al. (2020) showed DRO helps models learn invariant features. **Invariant Risk Minimization** (IRM) explicitly enforces that the model's feature-label relationship be the same across environments, discouraging reliance on environment-specific cues. Other methods add penalties: e.g. penalizing model reliance on a known feature (via adversarial confounder removal) or using entropy regularization to avoid high-confidence guessing. These often require knowledge of what the spurious attributes are (or splitting data by them) – a limitation in practice. As a citation, Chan et al. discuss ERM vs DRO, and Pathak et al. mention IRM and similar techniques in surveys ⁸ ²⁰.
- **Model and architecture changes:** Introducing inductive biases can preclude shortcuts. In vision, Geirhos et al. showed that training with stylized images makes CNNs focus on shape not texture ¹⁰. In NLP, restricting attention patterns or using structured reasoning modules might similarly ignore irrelevant cues. Some recent work uses *explainability-guided training*: e.g. one forces the model's attention or explanations to align with human-identified causal features, thereby discouraging shortcuts. Other ideas: ensembles of differently initialized models (if all agree despite input changes, it's likely robust) or feature disentanglement networks that separate style vs. content.

- **Evaluation protocols:** Finally, rigorous testing is itself a mitigation: by always including *anti-shortcut* test sets (like HANS for NLI or permutation tests for QA) in validation, one can catch shortcut-prone models early and tune them away. Measures like the **Internal Consistency Score** (do answers and explanations align?) or confidence calibration can flag overconfident shortcut behavior ⁹.

In practice, a combined approach works best. For example, Yuan et al. found that chain-of-thought (inference-time) *plus* careful choice of training data (e.g. including negation examples) drastically reduced error ⁴. Table 1 below summarizes some methods and examples:

Strategy	Category	Example Method / Reference	Notes
Data Augmentation	Data-centric	Counterfactual editing (insert/remove cues), back-translation, style transfer ¹⁰ ¹¹	Helps break specific spurious links. Requires realistic data generation.
Dataset Rebalancing	Data-centric	Upsample minority groups (WILDS style), filtering artifacts ¹⁶	Directly addresses bias but needs group labels.
Adversarial Training	Data-centric	Contrastive or adversarial examples targeting shortcuts (Qu et al., 2024)	Identifies and penalizes shortcut usage, but computationally costly.
Regularized Objectives (DRO/IRM)	Model-centric	Group DRO (Sagawa et al. 2020), IRM (Arjovsky et al. 2019)	Encourages invariance across environments. Requires env/group labels.
Optimizer / Batch Tuning	Training	Small-batch SGD or Adam (Marek et al. 2025) ⁷	Introduces noise, can escape sharp shortcut minima.
Prompt Engineering	Inference	Chain-of-thought (Yuan et al. 2024) ⁴ , majority voting	Reduces reliance on single-shot cues.
Interpretability-Guided Loss	Model-centric	LIME/SHAP-guided training (Kauffmann et al. 2025)	Penalizes attention to spurious features via explainability.
Evaluation Protocols	Other	Anti-shortcut test sets (HANS, permuted-MCQA)	Detects shortcut reliance so models can be iteratively fixed.

Experimental Recommendations

For practitioners wanting to **detect and mitigate shortcuts in LLMs**, we suggest the following protocols and settings:

Experiment	Tasks/Datasets	Metrics	Variations / Notes
Prompt-Shortcut Test	NLI (MultiNLI/SNLI + HANS ¹²), sentiment (SST2+triggers)	Accuracy drop; Internal Consistency (ICS) ⁹ ; Confidence Score	Vary model size (small→large), prompt type (zero-shot, few-shot, CoT) ⁴ . Insert/relocate shortcut tokens to measure sensitivity.
Batch-Size Sensitivity	Vision spurious tasks (Waterbirds, Colored MNIST ²¹) or language (biased text classification)	Worst-group vs. avg accuracy; training stability (loss fluctuations)	Fine-tune a smaller Transformer (e.g. GPT-2) on a spurious dataset. Compare batch sizes 1, 16, 128; compare optimizers (Adam vs SGD) ⁷ . Measure robustness of final model.
Counterfactual Augmentation	NLI or QA with known bias (e.g. include negated examples ¹¹)	ICS, Semantic Fidelity; drop in bias accuracy	Augment training data by reversing or scrambling the shortcut feature. Test whether model performance on bias vs. counterfactual inputs converges.
Debias Objective	WILDS tasks (CelebA, etc.)	Worst-group accuracy ⁸ ; average accuracy	Train with/without Group DRO or IRM. Perform ablation on penalty strength. Check for worst-case gains vs overall trade-off ²² .
Explanation Evaluation	Any QA or classification	Explanation coherence (EQS) ⁹ ; human evaluation	Generate model explanations. Use ICS/EQS to score them. Optionally verify explanations with human judgments.
Permutation/ Vocab Sensitivity	MCQA (MMLU, VQA)	Accuracy under random shuffles ¹⁴	Randomly shuffle answer choices or paraphrase queries. Observe accuracy changes. Test majority voting fix.
Ablation / Levenshtein Analysis	Benchmark tasks (HANS, etc.)	Change in answer after removing key tokens	Remove suspected shortcut tokens (e.g. “not”, proper nouns) and measure output.

For all experiments, report both *standard accuracy* and *robustness metrics* (e.g. worst-group accuracy ⁸ or “anti-shortcut” accuracy). Track model confidence on “trap” inputs vs. normal inputs. When training models, log how performance on in-distribution vs. biased splits evolves (to catch if one is improving at the expense of the other). Run ablations: e.g. compare zero-shot vs chain-of-thought prompts ⁴, or small vs large model sizes. As hyperparameters, one should try extreme settings highlighted by recent work: e.g. batch=1 and scaled learning rate ⁷, or long finetuning vs short, to see stability. In summary, experiments should systematically vary data, model, and training knobs and measure robustness (not just average accuracy) across groups and stress tests.

Open Problems and Future Directions

Despite progress, many questions remain. **Generalization of small-model insights to LLMs:** We still lack a unified theory of why some shortcuts diminish (Chan et al. 2022) while others grow (Tang et al. 2023) as models scale. Understanding how pretraining distributions and network scale interact with shortcut tendency is an open problem. **Unsupervised shortcut detection:** Current mitigation often relies on knowing the shortcut beforehand. Automated methods to **discover** spurious features (e.g. via counterfactual mining or unsupervised slicing) are needed (Pathak et al. emphasize this as future work ²⁰). **Evaluation benchmarks:** There is a shortage of comprehensive, *multi-domain* benchmarks for shortcut robustness. Most tests target one shortcut at a time (e.g. one spurious feature in one task). Creating large, realistic datasets with unknown spurious factors (and without group labels) is an open challenge ²². **Robust training trade-offs:** Many methods (DRO, IRM) improve worst-case performance but hurt average accuracy, and vice versa ²². Finding training objectives that avoid this trade-off – i.e. achieve both high average and worst-group accuracy – is an unsolved issue. **LLM-specific issues:** For generative LLMs, shortcuts can lead to subtle errors or “hallucinations” that are harder to quantify. Developing metrics that capture when an LLM is taking the easy way out (versus genuinely reasoning) is still nascent. **Scalability and interpretability:** Finally, as models grow (billions of parameters), existing debiasing techniques become harder to apply or interpret. New strategies that scale gracefully (e.g. using tiny fine-tuning modules like LoRA for debiasing) and tools to explain model decisions at scale are important research directions.

In summary, shortcut learning remains a pervasive challenge. We advocate continued development of *systematic tests* (datasets and metrics) and *algorithmic tools* (from data augmentation to robust training) in tandem. Iterative evaluation on known and newly discovered shortcuts should be standard practice. By following the experiments and suggestions above, researchers can benchmark and reduce spurious reliance in their LLMs. Achieving truly causal, generalizable language models is still an open frontier, calling for a mix of careful data design, clever training regimes, and rigorous evaluation.

¹ ² ¹⁷ [aclanthology.org](https://aclanthology.org/2023.findings-acl.284.pdf)

<https://aclanthology.org/2023.findings-acl.284.pdf>

³ ⁴ ⁵ ⁹ ¹² [aclanthology.org](https://aclanthology.org/2024.emnlp-main.679.pdf)

<https://aclanthology.org/2024.emnlp-main.679.pdf>

⁶ ¹⁴ [2310.01651] Fool Your (Vision and) Language Model with Embarrassingly Simple Permutations

<https://arxiv.org/html/2310.01651v3>

⁷ [2507.07101] Small Batch Size Training for Language Models: When Vanilla SGD Works, and Why Gradient Accumulation Is Wasteful

<https://arxiv.org/abs/2507.07101>

⁸ ¹³ ¹⁵ ²¹ ²² Spurious Correlations in Machine Learning: A Survey

<https://arxiv.org/html/2402.12715v2>

¹⁰ [1811.12231] ImageNet-trained CNNs are biased towards texture; increasing shape bias improves accuracy and robustness

<https://arxiv.org/abs/1811.12231>

¹¹ arxiv.org

<https://arxiv.org/pdf/2511.06234>

¹⁶ www2.eecs.berkeley.edu

<https://www2.eecs.berkeley.edu/Pubs/TechRpts/2022/EECS-2022-138.pdf>

¹⁸ ²⁰ **Frontiers | Unmasking the Clever Hans effect in AI models: shortcut learning, spurious correlations, and the path toward robust intelligence**

<https://www.frontiersin.org/journals/artificial-intelligence/articles/10.3389/frai.2025.1692454/full>

¹⁹ **On the Foundations of Shortcut Learning | OpenReview**

<https://openreview.net/forum?id=Tj3xLVuE9f>